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RESEARCH ARTICLE

Cluster Head-Based Cooperative Communication System for Efficient Energy Consumption in Swarm UAVs

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ABSTRACT Demand for unmanned aerial vehicles (UAVs) has surged across diverse domains. Because UAV systems are battery-powered, their available energy is limited, which in turn constrains transmit power and thus communication range. To mitigate these constraints, swarm UAVs—in which multiple UAVs cooperate to accomplish a mission—have attracted significant attention. In a swarm, multiple UAVs fly concurrently and perform tasks cooperatively; the operational area of each UAV is partitioned to minimize overlap. Consequently, swarms can systematically cover a wider area than a single UAV and increase mission throughput. However, in decentralized flying ad hoc network (FANET) environments, the large volume of UAV-to-UAV communications induces congestion, and mobility leads to high topological volatility, rendering links unstable. As a result, latency and packet loss may occur, making it difficult to guarantee quality of service (QoS). In centralized models proposed to address this issue, a cluster head manages communications. Because the cluster head consumes more energy than other UAVs, it has a shorter flight time. When the cluster head is replaced, the ensuing topology change degrades the overall network stability. To remedy the instability caused by rapid energy depletion of the cluster head, we propose a Cluster-Head-Based Cooperative Communications System (C3S) that leverages cooperative communications. C3S shares data with cooperative nodes and performs cooperative transmissions to reduce the energy consumption of the transmitting node, thereby balancing energy expenditure across cluster-head nodes and extending network lifetime. Moreover, C3S exploits temporal and spatial diversity gains to provide reliable communications at low transmit power.

INDEX TERMS UAV, cooperative communication, energy efficiency.

I. INTRODUCTION

Recently, demand for unmanned aerial vehicles (UAVs) has increased substantially, and UAVs are being deployed across diverse domains [1], [2]. Compared with traditional wireless sensor network (WSN) systems, UAVs offer much higher mobility. However, their limited battery capacity restricts

mission execution to a bounded operating range. To overcome the energy and coverage limitations of individual UAVs, swarm UAV systems, in which multiple UAVs cooperate on a single mission, have gained prominence. By distributing and parallelizing tasks among multiple UAVs, swarms can cover a wider area and reduce mission completion time [3]. This distributed processing paradigm improves overall system efficiency and enables complex missions that would be infeasible for a single UAV.

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Due to the UAV's high mobility, the topology of a swarm changes frequently. Moreover, being battery-powered, UAVs operate under strict energy constraints. Therefore, an energy-efficient communication architecture is essential for effective swarm operation. A representative UAV networking model is the flying ad hoc network (FANET). In a FANET, UAVs autonomously establish a network topology and determine routing paths without relying on external infrastructure, resulting in a highly dynamic wireless network [4], [5], [6]. Nodes communicate in a peer-to-peer manner without centralized control and often act as routers. However, due to rapid mobility, limited battery capacity, transmit power, and constrained bandwidth, FANET links frequently break and reform, which can degrade end-to-end network performance [7], [8].

To address these issues, centralized communication models have received increasing attention. In such models, a cluster head (CH) coordinates communications, reducing network management complexity and improving scalability through cluster-level control. Ordinary nodes communicate only with their CH, improving energy efficiency. Intra-cluster communications can be handled with simple routing, thereby reducing routing overhead within a cluster. However, because a CH processes and forwards both intra-cluster traffic and external communications (e.g., to the base station (BS)), it incurs substantial energy consumption, resulting in a shorter flight time than other UAVs. During a mission, once a CH exhausts its energy budget, it must abort its role and return to base, prompting the election of a new CH for that cluster. Frequent replacements, in turn, degrade overall network stability.

Therefore, to enhance the stability of centralized models, it is crucial to reduce the frequency of CH re-elections and mitigate excessive CH energy consumption. A CH typically expends the most energy while transmitting the cluster's data to the BS. Therefore, improving the communication process between the CH and the BS is essential for increasing the efficiency of centralized architectures. Existing studies [9], [10], [11], [12] primarily target wireless sensor networks (WSNs) and focus on energy-efficient routing protocols and data transmission algorithms. However, because these approaches are primarily designed for static sensing environments, their direct applicability to UAV networks, characterized by high mobility and rapidly changing topology, is limited.

To address energy consumption in swarm UAV networks, we propose a Cluster-Head-Based Cooperative Communications System (C3S) that applies cooperative communications to a centralized model [13], [14]. C3S dynamically evaluates each UAV's channel quality and energy consumption and enables CHs to transmit cooperatively. By doing so, C3S reduces the CH's energy burden, a key weakness of centralized designs, thereby improving overall network performance and extending network lifetime.

The remainder of this paper is organized as follows. Section II reviews prior work on cooperative communications in swarm UAV networks and discusses their limitations

in highly dynamic UAV settings. Section III presents the proposed C3S architecture. Section IV reports the experimental setup and performance evaluation. Finally, Section V concludes the paper.

II. RELATED WORK

In the multi-cluster-head centralized model [15], multiple cluster heads (CHs) coexist. Each CH is responsible only for its own region and can therefore respond independently to local changes. When a new UAV is added, it is associated with the nearest cluster, thereby limiting the impact to that cluster and ensuring optimal scalability. Additionally, inter-cluster communications can proceed in parallel, thereby reducing the overall communication load of the network. This architecture effectively manages complex swarm-UAV systems, as illustrated in Fig. 1, and enables cluster-level control in complex environments where multiple UAVs operate concurrently. Each UAV collects diverse sensor data, reacts to its environment in real-time, and may perform multiple tasks simultaneously. In this model, each CH UAV manages communications and mission coordination within its assigned cluster. Among the CHs, one acts as a *master cluster head* (MCH), supervising the set of CHs and relaying communications between them and the base station (BS). This hierarchical approach distributes network load across multiple CHs, thereby allowing each CH to administer an independent area. This, in turn, increases management efficiency in large-scale operations.

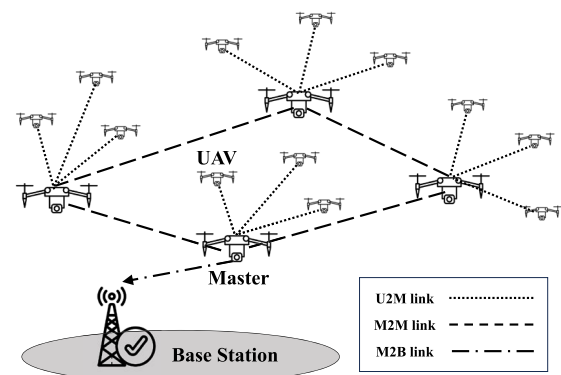


FIGURE 1. Multi-cluster head centralized network.

A CH collects data from the member nodes in its cluster and forwards it to the BS or to the MCH, thereby directly affecting overall network performance and efficiency. Because the CH consumes more energy than other nodes during data collection, aggregation, and forwarding, electing an energy-inefficient node as the CH may lead to rapid energy depletion and frequent CH replacement [16], [17]. Given the high mobility of UAVs and the resulting dynamic topology, selecting a node located at a stable position is also important. If a node with frequent position changes or poor link quality becomes the CH, intra-cluster reliability degrades and re-clustering occurs frequently [18]. As the frequency of CH re-election increases, network

stability and efficiency decline significantly. Therefore, effective clustering in UAV networks critically depends on appropriate CH selection [19]. Representative CH-selection and clustering methods include LEACH (Low-Energy Adaptive Clustering Hierarchy) [17] and k-means [18]. LEACH is designed to maximize energy efficiency by randomly rotating the CH role to balance energy consumption and extend network lifetime. Its two-phase protocol selects CHs during the setup phase and performs data transmission and management during the steady-state phase, iterating through rounds so that nodes share the CH role fairly. k-means groups nodes into k clusters based on proximity to cluster centroids. The node closest to a centroid is chosen as the CH. By forming clusters that minimize the sum of distances between nodes and their CH, k-means can improve communication efficiency and determine an optimal cluster structure and CHs. However, because these methods are primarily designed under static-environment assumptions, they may fail to perform well in highly dynamic UAV settings. This motivates dynamic clustering approaches that include real-time link quality, node energy state, and changes in UAV position.

In dynamic environments, beyond clustering, additional techniques are required to maintain robust communications. Cooperative-communication-based studies for reliability improvement [20] employ relay nodes to mitigate path loss and leverage diversity techniques to optimize the signal-to-noise ratio (SNR), thereby extending coverage and enhancing link stability. Based on cooperative communications, the C³ RUN (Cooperative Communication Based Connectivity Recovery algorithm for UAV Networks) [21] has been proposed to address connectivity loss and cluster partitioning in UAV networks. The efficacy of cooperative transmission is determined by the selection of high-quality cooperative nodes, for which link quality is a crucial metric. In UAV networks, link quality can fluctuate rapidly due to mobility and channel variability, making it crucial to accurately assess and incorporate link-quality metrics in a timely manner.

Link quality is commonly evaluated using SNR, as expressed in (1). In (1), Original SNR_{*i,j*} denotes the SNR of the link between node i and node j . SNR quantifies the ratio of received signal power to noise power; a higher SNR indicates a higher-quality link. In mobile wireless systems, an SNR of 10 dB or higher is typically considered sufficient for reliable communication [22]. Here, $P_{i,j}$ is the transmit power from node i to node j ; $h_{i,j}$ is the channel gain between them. Noise is the noise power in the network; $d_{i,j}$ is the distance between nodes i and j ; and α is the path-loss exponent determined by the propagation environment. Equation (2) removes constant-factor terms such as 10, log, $h_{i,j}$, and Noise from (1). Because these terms are treated as constants, link quality is evaluated using (2), which retains only the variable terms.

$$\text{Original SNR}_{i,j} = 10 \cdot \log \frac{P_{i,j} \cdot |h_{i,j}|^2}{\text{Noise} \cdot d_{i,j}^\alpha} \quad (1)$$

$$\text{SNR}_{i,j} = \frac{P_{i,j}}{d_{i,j}^\alpha} \quad (2)$$

When all nodes u in the UAV set U cooperatively transmit data to the base station bs, the SNR is given by (3) [23].

$$\text{SNR}_{\text{coop}} = 10 \cdot \log \sum_{u \in U} \frac{\frac{P_{u,bs}}{|U|} \cdot |h_{u,bs}|^2}{\text{Noise} \cdot d_{u,bs}^\alpha} \quad (3)$$

Equation (3) represents the SNR under cooperative transmission among UAVs. When $|U|$ nodes cooperate, each node can reduce its transmit power by a factor of $1/|U|$ while achieving the same SNR as single-node transmission. Data received from different cooperating nodes at different times provides spatiotemporal diversity; combining such diverse signals produces higher-quality data even under the same SNR [23]. This diversity gain plays a critical role in ensuring reliability, particularly when the main transmitting node is too distant from the BS to achieve the target SNR via direct transmission. C3S can overcome this limitation by combining signals from multiple nodes. Thus, cooperative communications improve not only energy efficiency but also the reliability of the received data.

Cluster-based network improves energy efficiency and stability in UAV systems by enabling cluster-level communication and management; however, they still lack flexibility in dynamic environments and tend to cause CH overload. A failure at the CH can act as a single point of failure, impacting the entire network. Cooperative communication techniques are effective in mitigating path loss and optimizing SNR to enhance link stability, and SNR-based link-quality evaluation is a key factor in assessing communication reliability and efficiency. However, these existing approaches do not resolve the energy consumption problem in CH UAV networks. To overcome this limitation, research is needed to distribute the CH's transmit power efficiently without compromising communication reliability.

Recently, the Drone-Assisted Cooperative Routing (DACR) technique has been proposed, which utilizes drones as auxiliary relays in Vehicle-to-Everything (V2X) communication environments [24]. In this approach, if communication between ground vehicles fails due to congestion or link disconnection, a drone acts as an intermediate relay node to restore communication. This process uses a composite routing metric that considers direction, speed, and traffic load to select the optimal next-hop vehicle, thereby reducing overhead and latency. This method leverages UAVs as relays to ensure network robustness and seamless connectivity. However, DACR focuses solely on connectivity and does not address high energy consumption or load-balancing issues among nodes.

Research utilizing Reconfigurable Intelligent Surfaces (RIS) as a new hardware alternative to active relays to improve the link quality and energy efficiency of UAV communications has also gained attention [25]. RIS employs a programmable array of reflective elements to create a virtual Line-of-Sight (LoS) link, enhancing communication

quality even in environments where the signal is obstructed. By comparing the performance of traditional active Decode-and-Forward (DF) relays and RIS, simulations have shown that at high altitudes (above 100 m), RIS can achieve superior performance with less power than relays. While RIS can provide indirect power gains by improving link quality, it does not address the problem of traffic and load concentration on specific nodes.

Beyond the communication layer cooperation, recent research has proposed a framework where a UAV swarm is responsible for wireless power transfer to charge low-power IoT devices [26]. To ensure the swarm's sustainability, this model also includes charging stations (CSs) that can recharge the UAVs themselves. To manage these complex interactions, matching theory was employed to orchestrate the swarm and establish stable, incentive-based linkages between UAVs and CSs. This energy supply and charging orchestration addresses a complementary problem to that solved by C3S. Whereas these frameworks focus on the high-level logistics of energy supply and recharging, C3S focuses on the energy efficiency of the data communication process during mission execution. The cooperative transmission mechanism of C3S can be integrated into such matching-based orchestration frameworks to further enhance the swarm's overall energy efficiency.

III. CLUSTER HEAD-BASED COOPERATIVE COMMUNICATION SYSTEM

This section describes the Cluster-Head-Based Cooperative Communications System (C3S), a cooperative transmission framework for energy-efficient swarm-UAV operations. C3S is a distributed transmission model in which the main transmitting node collaborates with other cluster heads to deliver data. By allowing cooperative transmission, the main transmitting node can operate at a lower transmit power than in single-node transmission, thereby relieving the cluster-head overload that is a key drawback of centralized architectures.

A. ARCHITECTURE

In C3S, UAVs play roles as cluster heads and cluster members. A cluster head (CH) may act either as the main transmitting node or as a cooperating node that assists the transmission. A CH processes data within its cluster and forwards it to other CHs or to the base station (BS). A cluster member is a UAV belonging to a cluster; it performs mission tasks, collects data, and forwards the data to its CH. Cluster members move dynamically within the network and may be elected as CHs. C3S does not directly address the clustering or CH selection algorithms themselves. It assumes that clustering and CH selection have already been completed by an underlying protocol, such as LEACH or k-means.

When a particular CH intends to transmit data to the BS, it is referred to as the *main transmitting node*. The nodes that assist this transmission are called *cooperative nodes* and are selected from among the CHs. Cooperative-node

selection considers the channel states of the links between a cooperative node and the main transmitting node, between cooperative nodes, as well as between the cooperative node and the BS, together with node positions. Fig. 2 illustrates the role composition of nodes in the network. By default, all nodes start as cluster members and may be promoted to CHs depending on network conditions and node status. When a CH needs to send the data to the BS, it becomes the main transmitting node, and some of the other CHs support the transmission as cooperative nodes. This hierarchical and flexible role assignment improves both efficiency and stability.

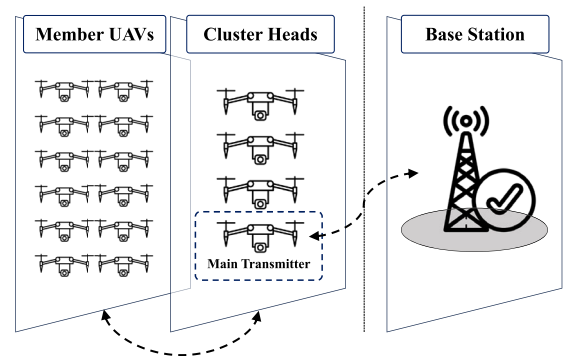


FIGURE 2. Role composition of C3S.

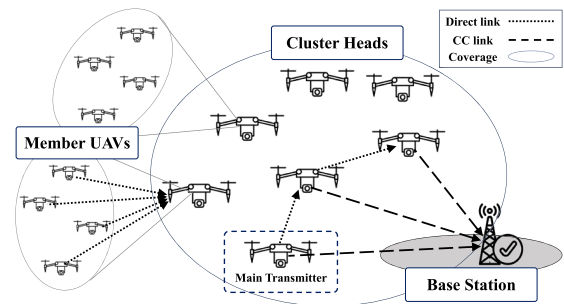


FIGURE 3. Communication structure model of C3S.

As shown in Fig. 3, C3S employs two types of links: the *direct link* and the *cooperative-communication (CC) link*. The direct link is used for transmissions between cluster members and their CH, or between CHs. The CC link is used when multiple cooperative nodes jointly forward the data to the BS. This communication model evaluates performance using the signal-to-noise ratio (SNR) and maintains an SNR of at least 10 dB to ensure reliable data delivery.

Before the transmission, the main transmitting node selects cooperative nodes, shares the data with them, and then communicates with the BS using cooperative transmission. This enables reliable BS connectivity at a lower transmit power than single-node transmission.

This paper focuses on the design and analysis of a cooperative transmission algorithm to enhance the energy efficiency of UAV networks. To analyze the pure energy gains of the C3S algorithm itself, this system model assumes an ideal channel environment. The channel model considers only

path loss in Free-Space and excludes complex non-linear channel effects such as multipath fading or the Doppler effect. It is also assumed that no packet collisions or interference occur. Furthermore, the energy consumption model in this paper focuses only on transmission power to evaluate the algorithmic efficiency of the C3S protocol. In terms of mobility, C3S assumes a quasi-static environment. This is to isolate the intrinsic performance of the algorithm, excluding the effects of rapid topological changes caused by high mobility.

B. COOPERATIVE NODE SELECTION

Before the data transmission, C3S selects cooperating nodes. We first compute the total transmit power required to deliver the data and then choose the optimal set of cooperative nodes accordingly. The node that intends to transmit the data is referred to as the *main transmitting node* and is denoted by m , while the receiving base station is denoted by bs . To guarantee communication reliability during the selection process, all links are configured to satisfy the 10 dB SNR target in (1). Let $P_{i,j}$ denote the transmit power that achieves 10 dB SNR for the link from node i to node j ; transmitting with $P_{i,j}$ ensures reliable communication on link $i \rightarrow j$.

In C3S, the total transmit power required for m to deliver data to bs at 10 dB SNR is denoted by P_{total} . As shown in (4), P_{total} is the sum of the share power P_{share} , used to disseminate the data among the cooperative nodes, and the cooperative-transmission power P_{coop} , used to forward the data to the BS.

$$P_{\text{total}} = P_{\text{share}} + P_{\text{coop}} \quad (4)$$

In Equation (5), H is the set of cluster heads, S the set of cooperative nodes selected for cooperative transmission, and C the set of candidate nodes that can become cooperative nodes. When initially searching for cooperative nodes, there are no cooperative nodes and only the main transmitting node exists, so the process starts with $S = \{m\}$. At each step, a new cooperative node is sought from C and, if selected, is added to S . This iteration continues until no candidate can further reduce the total power. Let $N = |S|$ denotes the current cooperation size. P_{total_N} is the total transmission power when $|S|=N$, i.e., when N nodes perform cooperative transmission. The first term accounts for sharing the data along the chain $s_1 \rightarrow s_2 \rightarrow \dots \rightarrow s_N$, while the second term represents equal-power cooperative transmission from all nodes in S to the BS. $P_{\text{total}_N}(c)$ is the total transmission power when node c is added as a cooperative node to P_{total_N} .

$$C = \{c_1, c_2, \dots\}, \text{ where } C = H \setminus S \quad (5)$$

$$P_{\text{total}_N} = \sum_{i=1}^{N-1} P_{s_i, s_{i+1}} + \sum_{i=1}^N \frac{P_{s_i, bs}}{N} \quad (6)$$

$$P_{\text{total}_N}(c) = \sum_{i=1}^{N-1} P_{s_i, s_{i+1}} + P_{s_N, c} + \frac{P_{c, bs}}{N+1} + \sum_{i=1}^N \frac{P_{s_i, bs}}{N+1} \quad (7)$$

The cooperative node search process starts with $S = \{m\}$, then find c^* among all $c \in C$ that minimizes $P_{\text{total}_N}(c^*)$. If $P_{\text{total}_N}(c^*)$ is not smaller than P_{total_N} , the procedure terminates; otherwise, c^* is added to S and the iteration continues, with $C \leftarrow H \setminus S$. This process is expressed by Equation (8) and represented algorithmically as Algorithm 1.

$$P_{\text{total}_{N+1}} = P_{\text{total}_N}(c^*),$$

$$\text{where } c^* = \arg \min_{c \in C} P_{\text{total}_N}(c), P_{\text{total}_N}(c^*) \leq P_{\text{total}_N} \quad (8)$$

Algorithm 1 Dynamic Cooperative Cluster Head Selection

- 1: **Initial Setup:**
- 2: Identify all cluster head nodes and base station in network
- 3: **Input:** Main transmitting node m , set of CHs H , base station bs
- 4: **Output:** Set of nodes participating in cooperative transmission S
- 5: $S \leftarrow \{m\}$ // initialize selected set
- 6: $C \leftarrow H \setminus S$ // candidate set for cooperative nodes
- 7: **while** $|C| > 0$ **do**
- 8: $N \leftarrow |S|$
- 9: $c^* \leftarrow \arg \min_{c \in C} P_{\text{total}_N}(c)$
- 10: **if** $P_{\text{total}_N}(c^*) > P_{\text{total}_N}$ **then**
- 11: **return** S
- 12: **end if**
- 13: $S \leftarrow S \cup \{c^*\}$
- 14: $C \leftarrow H - S$
- 15: **end while**
- 16: **return** S

Through this process, the final cooperating set S is obtained, and it is used for the cooperative transmission phase. The total power consumption P_{total} used at this time consists of two parts as shown in Equation (9). The term $\sum_{i=1}^{N-1} P_{s_i, s_{i+1}}$ represents the sum of transmission power consumed for sharing data to be transmitted to bs among cooperative nodes. The term $\sum_{i=1}^N \frac{P_{s_i, bs}}{N}$ represents the average power consumed when cooperative nodes cooperatively transmit data to bs .

$$P_{\text{total}} = \sum_{i=1}^{N-1} P_{s_i, s_{i+1}} + \sum_{i=1}^N \frac{P_{s_i, bs}}{N}$$

$$= \sum_{i=1}^{N-1} P_{s_i, s_{i+1}} + \text{Average}(P_{s, bs}) \mid s \in S, N = |S| \quad (9)$$

This selection mechanism constructs an energy-efficient set of cooperative nodes, providing clear benefits in terms of energy consumption and network lifetime, while enabling stable cooperative communications in realistic wireless sensor network deployments.

C. ANALYTICAL ENERGY MODEL

The CH collects data from its cluster members and transmits it to the BS. We analyze the impact of the number of cluster

members, k , on the energy required for the CH to transmit to the BS. When each cluster member intends to send B bits of data, the total data size D that the CH must transmit to the BS is $k \times B$ bits. The time t required for the CH to send data D is $\frac{D}{R}$, where R is the data rate.

$$\begin{aligned} E_{\text{dir}}(k) &= P_{m,\text{bs}} \times t \\ &= P_{m,\text{bs}} \times \left(\frac{D}{R}\right) \\ &= P_{m,\text{bs}} \times \frac{k \times B}{R} \\ &= \frac{k \times B}{R} \times P_{\text{total}_1} \end{aligned} \quad (10)$$

$$\begin{aligned} E_{\text{C3S}}(k) &= P_{\text{share}} \times t_{\text{share}} + P_{\text{coop}} \times t_{\text{coop}} \\ &= \left(\sum_{i=1}^{N-1} P_{s_i, s_{i+1}} \times t\right) + \left(\sum_{i=1}^N \frac{P_{s_i, \text{bs}}}{N} \times t\right) \\ &= \left(\sum_{i=1}^{N-1} P_{s_i, s_{i+1}} \times \frac{k \times B}{R}\right) \\ &\quad + \left(\sum_{i=1}^N \frac{P_{s_i, \text{bs}}}{N} \times \frac{k \times B}{R}\right) \\ &= \frac{k \times B}{R} \times \left(\sum_{i=1}^{N-1} P_{s_i, s_{i+1}} + \sum_{i=1}^N \frac{P_{s_i, \text{bs}}}{N}\right) \\ &= \frac{k \times B}{R} \times P_{\text{total}_N} \end{aligned} \quad (11)$$

The total energy used for direct transmission $E_{\text{dir}}(k)$ and C3S cooperative transmission $E_{\text{C3S}}(k)$, when there are k cluster members, is defined as (10), (11). From Algorithm 1, $P_{\text{total}_1} \geq P_{\text{total}_N}$ which implies $E_{\text{dir}}(k) \geq E_{\text{C3S}}(k)$. Therefore, the total energy used for transmission in C3S is always less than or equal to that of the Direct method.

$$\begin{aligned} E_{\text{dir,per-node}}(k) &= \frac{E_{\text{dir}}(k)}{1} \\ &= \frac{k \times B}{R} \times P_{\text{total}_1} \end{aligned} \quad (12)$$

$$\begin{aligned} E_{\text{C3S,per-node}}(k) &= \frac{E_{\text{C3S}}(k)}{N} \\ &= \frac{k \times B}{R} \times \frac{P_{\text{total}_N}}{N} \end{aligned} \quad (13)$$

We also examine the average energy per node, $E_{\text{per-node}}$. As shown in (12) and (13), $E_{\text{per-node}}$ is the total energy divided by the number of CHs participating in the transmission. The Direct method has one transmitting CH, so its per-node energy equals the total energy. C3S distributes the energy among N cooperative nodes. Thus, the per-node energy efficiency of C3S improves as total energy and N increase.

$$\begin{aligned} E_{\text{dir,per-bit}}(k) &= \frac{E_{\text{dir}}(k)}{D} \\ &= \frac{E_{\text{dir}}(k)}{k \times B} \\ &= \frac{P_{\text{total}_1}}{R} \end{aligned} \quad (14)$$

$$\begin{aligned} E_{\text{C3S,per-bit}}(k) &= \frac{E_{\text{C3S}}(k)}{D} \\ &= \frac{E_{\text{C3S}}(k)}{k \times B} \\ &= \frac{P_{\text{total}_N}}{R} \end{aligned} \quad (15)$$

The per-bit energy, $E_{\text{per-bit}}$, shows a different trend. As shown in (14) and (15), $E_{\text{per-bit}}$ is the total transmission energy divided by the data size D . The per-bit energy is not affected by the number of cluster members k ; it is only influenced by the transmission power and data rate.

D. OVERHEAD

While C3S provides significant energy savings, it introduces overhead in two areas. The first is the computational overhead required to select the nodes for cooperative transmission. The second is the latency overhead incurred from sharing the data among cooperative nodes before the joint transmission to the BS. Before cooperation begins, the main transmitting node m selects the set of cooperative nodes S . Algorithm 1 finds a cooperative node by iterating through all candidates and calculating the total transmission power for each potential addition. This iteration continues until no more nodes can be added or no further energy gains are found, which is proportional to the number of cluster head nodes $|H|$. Furthermore, the argmin operation to select the node that maximizes efficiency is performed for all $c \in C$. Since the size of C is at most $|H|$, the overall time complexity of Algorithm 1 is $O(|H|^2)$.

To perform cooperative transmission, the selected cooperative nodes must first share the data to be transmitted. As shown in the first term of (6), C3S involves sequential data sharing among cooperative nodes before joint transmission to the BS. This process incurs a latency of $O(|S|)$, proportional to the number of cooperative nodes $|S|$. C3S presents a clear trade-off between energy efficiency and transmission latency. Cooperating with more nodes improves the energy efficiency of cooperative transmission, but it incurs $O(|S|)$ latency for data sharing. An increasing data-sharing time is problematic, as the cooperative nodes may move during this period, altering the topology. If the data-sharing time is too long, the topology will differ from when the set S was calculated, rendering the selection suboptimal or even invalidating the link quality assumptions. In high-mobility environments, frequent topological changes are unavoidable, so this data-sharing time must be minimized to ensure the algorithm's validity. Therefore, it is crucial to balance energy efficiency and transmission latency by setting an appropriate $|S|$ based on the mission environment.

IV. PERFORMANCE EVALUATION

This section evaluates the performance of the proposed C3S using MATLAB-based simulations. We first describe the simulation environment and parameters, and then analyze energy-consumption patterns under direct and cooperative transmission scenarios.

A. SIMULATION ENVIRONMENTS

For the performance evaluation of C3S, we use MATLAB, which provides numerical tools and visualization capabilities well-suited for system modeling and simulation. The communication protocol follows the widely adopted IEEE 802.11 standard, and the 2.4 GHz band is used to ensure robust connectivity. IEEE 802.11 offers broad compatibility and stability, and is widely employed in WSNs and swarm-UAV networks [19], [27]. The 2.4 GHz band provides favorable propagation and obstacle penetration characteristics for UAV communications.

The network space is configured as $250\text{ m} \times 250\text{ m} \times 250\text{ m}$, reflecting the typical 250–300 m range where stable links can be maintained in the 2.4 GHz band under the 802.11 standard. This setting enables all nodes to exchange data effectively throughout the volume. The base station (BS) is placed at $(0, 0, 0)$.

The data rate is set according to IEEE 802.11n in the 2.4 GHz band. With a 20 MHz channel and a baseline modulation and coding scheme (MCS), the nominal maximum PHY rate is set to 72.2 Mb/s. We employ the A-MPDU (aggregated MAC protocol data unit) frame aggregation technique available in IEEE 802.11n and beyond, and configure the payload size from 4 KB to 64 KB [28]. The maximum 64 KB setting approaches the theoretical upper limit of A-MPDU aggregation to fully leverage frame-aggregation gains.

Time Division Multiple Access (TDMA) is used as the channel-access method. Each node is assigned a dedicated time slot and transmits periodically. A time slot consists of the actual data-transmission time and a guard time for collision avoidance. The packet transmission time is computed by dividing the packet size by the data rate. For the maximum packet size of 64 KB at 72.2 Mb/s, the transmission time is 7.26 ms. Adding a 0.74 ms guard time to mitigate inter-slot interference and avoid collisions yields a final slot duration of 8 ms. Since one time slot is 8 ms, there are 125 slots per second, allowing at most 125 nodes to be scheduled concurrently in the network.

The channel model in the simulation environment follows the Free-Space model; therefore, the path-loss exponent is set to the theoretical value of 2. Furthermore, all power values are presented as unitless relative values, normalized with $\text{Noise} = 1$ and $\text{Channel Gain} = 1$. Therefore, $P_{i,j}$ is calculated as $P_{i,j} = 10 \times d_{i,j}^2$. To ensure reliable communications, the receiver SNR threshold is set to 10 dB, reflecting a minimum requirement to suppress packet loss in practical wireless deployments.

Simulation parameters are summarized in Table 1.

B. SIMULATION SCENARIOS AND PERFORMANCE COMPARISON

Simulations are conducted with varying numbers of cluster heads, and nodes are placed using the k-means algorithm. Under the assumed TDMA environment, the maximum

TABLE 1. Simulation parameters.

Parameter	Value
Network Space	$250\text{ m} \times 250\text{ m} \times 250\text{ m}$
Base Station (BS) Location	$(0, 0, 0)$
Communication Protocol	IEEE 802.11n / 2.4 GHz
Payload Size	4 KB – 64 KB
Max Packet Size	64 KB
Channel Access Method	TDMA
Transmission Time	7.26 ms
Guard Time	0.74 ms
Final Time Slot Length	8 ms
Slots per Second	125
Path Loss Exponent	2 (Free-Space)
Receiver SNR Threshold	10 dB
Noise Power	1 (Normalized)
Channel Gain	1 (Normalized, Free-Space)
Total Number of Nodes (UAVs)	125
Node Deployment Method	k-means
Number of Cluster Heads Range	1 – 60
Minimum Cluster Size	2 (1 CH, 1 member node)

number of UAVs that can be scheduled concurrently is 125. To evaluate the system across diverse network topologies, we repeat node placement 10,000 times for each configuration, where the number of cluster heads ranges from 1 to 60 out of the 125 total nodes. We then calculate the average energy consumption to compare performance. Because each cluster must contain at least one cluster head and one ordinary node, the minimum cluster size is two. Given 125 total nodes, the maximum number of clusters is 62; therefore, we limit the number of cluster heads to at most 60.

Fig. 4 illustrates the network structure for direct transmission with a single cluster head. All nodes form a star topology by communicating directly with the central cluster head, and only that cluster head communicates with the base station (BS). In this topology, traffic concentrates on the cluster head, and nodes located near the network boundary experience longer transmission distances, leading to increased energy consumption.

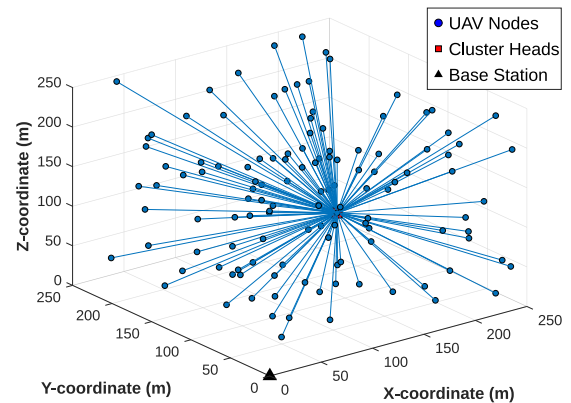


FIGURE 4. Visualization of single cluster head communication model.

Fig. 5 shows the cooperative-communication structures for multiple cluster heads (5, 10, 15, and 20). Each node connects to its nearest cluster head, and the cluster heads cooperate

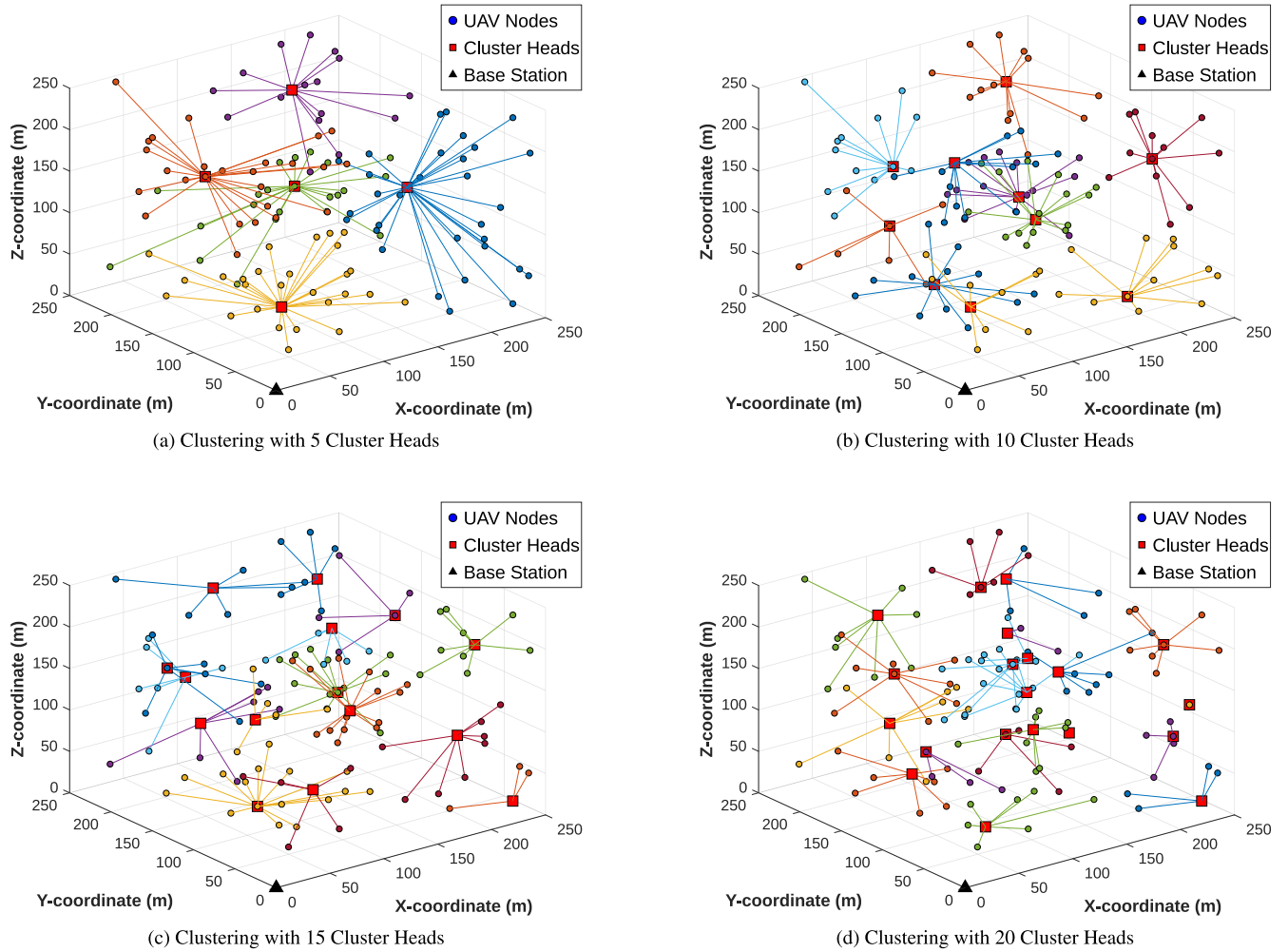


FIGURE 5. Visualization of multiple cluster head communication model.

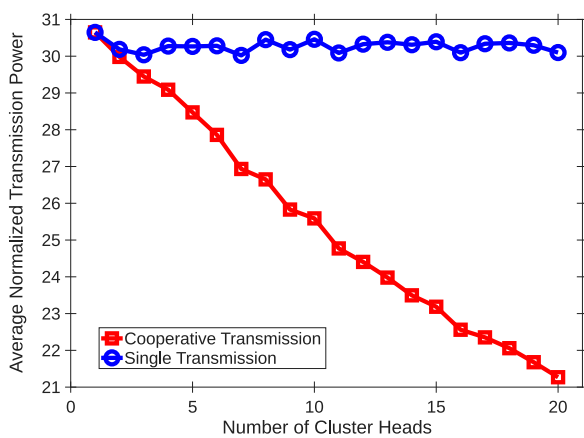


FIGURE 6. Comparison of average energy consumption.

to forward data to the BS. As the number of cluster heads increases, the network load is distributed, and the average transmission distance per node decreases.

Fig. 6 compares energy-consumption patterns between cooperative and direct transmission. We set the Direct

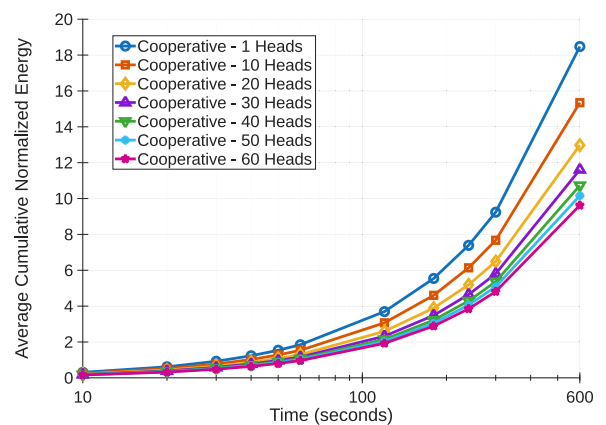


FIGURE 7. Average energy consumption over time with varying number of cluster heads.

Transmission method, which is the communication method used in representative cluster-based protocols such as LEACH, as the baseline. In the direct case, because a single main transmitting node sends data directly to the BS regardless of the number of cluster heads, energy consumption

TABLE 2. Average energy consumption and standard deviation between direct and cooperative transmission by number of cluster heads.

Cluster Heads	Direct Trans.		Cooperative Trans.	
	Avg.	Std. Dev.	Avg.	Std. Dev.
1	30.322	21.341	30.322	21.341
2	30.157	21.464	29.928	21.201
3	30.359	21.265	29.715	20.520
4	30.261	21.113	29.072	19.742
5	29.974	21.344	28.131	19.202
6	30.459	21.250	27.905	18.348
7	30.165	21.121	27.036	17.649
8	30.072	21.058	26.376	17.029
9	29.834	21.125	25.521	16.486
10	30.537	21.446	25.497	16.155
11	29.970	21.052	24.648	15.474
12	30.155	21.280	24.196	15.109
13	30.398	21.435	23.936	14.801
14	29.964	21.034	23.199	14.186
15	30.207	21.053	22.968	13.907
16	30.501	21.354	22.739	13.744
17	30.369	21.216	22.370	13.408
18	30.369	21.362	21.893	13.158
19	30.235	21.124	21.570	12.818
20	30.510	21.323	21.373	12.650

remains nearly constant and relatively high at approximately 3×10^5 J. This is because each node must communicate over the full distance to the BS via the central head, leading to distance-proportional energy consumption. In contrast, under cooperative transmission, energy consumption continuously decreases as the number of cluster heads increases. With a single cluster head, energy is comparable to the direct case; when the number increases to 20, energy drops to approximately 2.1×10^5 J, reducing it by about 30% compared to direct transmission. Cooperative transmission distributes the transmission burden among multiple nodes, thereby reducing the transmit power per node and improving energy efficiency. This reduction becomes more noticeable as the number of cluster heads grows.

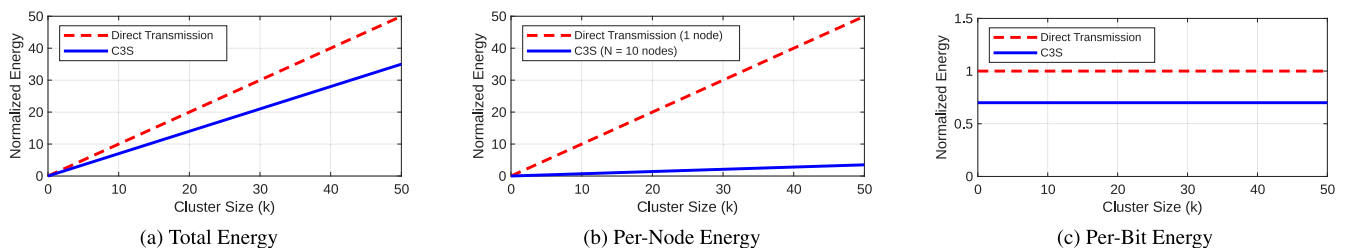
Table 2 compares the average energy consumption and standard deviation for direct and cooperative transmission as a function of the number of cluster heads. For 1 to 20 cluster heads, the direct-transmission case exhibits an average standard deviation of 21.426, whereas the cooperative case shows 18.719. The larger variability in the direct case indicates more fluctuation in node-level energy usage. Notably, cooperative transmission achieves a 12.63% lower standard deviation than direct transmission, providing more

stable and uniform energy consumption across the network. This not only reduces the average energy but also balances node-wise energy usage, thereby extending network lifetime.

Fig. 7 presents energy consumption over time with cluster head numbers of 1, 10, 20, 30, 40, 50, and 60. As the number of cluster heads increases, cooperative transmission becomes more energy-efficient, while the total energy consumption increases exponentially over time. The reduction rates with respect to the number of cluster heads are approximately 16.39% (1 to 10), 15.68% (10 to 20), 10.50% (20 to 30), 7.57% (30 to 40), 6.14% (40 to 50), and 5.12% (50 to 60), indicating diminishing returns as the number of cluster heads grows. A gradual decrease follows the initial rapid reduction because the additional benefits of load distribution and average link-length reduction become smaller as more cluster heads are introduced. This gradual slowdown in the rate of energy reduction suggests that the energy gains achievable through cooperative communication are converging to a limit, implying a theoretical lower bound on energy consumption.

These results demonstrate that C3S outperforms direct transmission in terms of energy efficiency and stability in UAV networks by leveraging cooperative communications. Moreover, given the diminishing returns observed as the number of cluster heads increases, selecting an appropriate number of cluster heads that matches network scale and operational objectives is critical to maximizing C3S performance.

Fig. 8 shows the energy scaling with the number of cluster members, k , from the model in Section III-C. To more clearly verify the energy change according to the number of cluster members k , we set $B = 1$, $R = 1$, and $P_{\text{total}_1} = 1$ to normalize all energy values relative to the energy of Direct Transmission, and set $N = 10$. Based on previous experimental results showing that cooperative transmission reduces transmission power by 30%, we set $P_{\text{total}_N} = 0.7 \times P_{\text{total}_1}$. In Fig. 8a, the total energy is linearly proportional to k . Since $P_{\text{total}_N} < P_{\text{total}_1}$, C3S always shows a lower slope, which means better energy efficiency than the Direct method. In Fig. 8b, the per-node average energy is compared. It clearly shows that C3S effectively distributes the load across N nodes, resulting in significantly lower scaling than the Direct method. Finally, Fig. 8c theoretically validates that the per-bit energy is constant and independent of k , and that C3S achieves higher per-bit energy efficiency regardless of changes in k .

**FIGURE 8.** Theoretical per-transmission energy scaling based on the analytical model.

V. CONCLUSION AND FUTURE WORK

This paper proposed a Cluster-Head-Based Cooperative Communications System (C3S) for energy-efficient communications in swarm UAV networks. To assess its effectiveness, we analyzed energy-consumption patterns under varying numbers of cluster heads and compared cooperative transmission with direct transmission. The evaluation shows that cooperative transmission achieves a reduction of up to 30% in energy consumption compared to direct transmission. We also observed a gradual improvement in energy efficiency as the number of cluster heads increased. Additionally, cooperative transmission exhibits a 12.63% lower standard deviation of energy consumption compared to direct transmission, indicating more uniform node-level energy usage across the network. An analysis of the reduction rate further revealed diminishing returns as the number of cluster heads grows. Future work will pursue multi-objective optimization that jointly considers not only energy efficiency but also latency and throughput, enabling effective operation in practical swarm-UAV missions that demand real-time control responsiveness and high-volume data delivery (e.g., high-definition video).

As UAV mobility performance improves, we will conduct research to apply C3S in high-mobility environments by introducing link quality prediction techniques based on UAV movement and dynamically adjusting the number of cooperative nodes according to the rate of topological change. We will also conduct experiments that incorporate various non-linear effects, interference, and packet collisions that inevitably occur in real-world environments into the C3S cooperative transmission model to quantitatively evaluate its energy-saving benefits and robustness in more realistic settings. Furthermore, the current energy consumption model considers only transmission-related power to isolate the algorithmic advantages of the C3S protocol. However, the dominant components of UAV energy consumption in practical deployments are propulsion, hovering, and payload-related costs. Future work will investigate advanced models that explicitly integrate the complex relationships between propulsion dynamics, communication payload, and the UAV's battery life and optimal trajectory.

The energy efficiency and communication reliability improvements provided by C3S are critical for practical missions. For example, in disaster response scenarios, C3S can provide a robust ad-hoc communication network. In continuous surveillance missions, it enables longer-duration, energy-efficient transmission of high-definition video. It can also play a key role in supporting long-duration data collection missions in large-scale, IoT-based aerial networks.

REFERENCES

- [1] M. Ghamari, P. Rangel, M. Mehrubeoglu, G. S. Tewolde, and R. S. Sherratt, "Unmanned aerial vehicle communications for civil applications: A review," *IEEE Access*, vol. 10, pp. 102492–102531, 2022.

- [2] H. Shakhatreh, A. H. Sawalmeh, A. Al-Fuqaha, Z. Dou, E. Almaita, I. Khalil, N. S. Othman, A. Khreishah, and M. Guizani, "Unmanned aerial vehicles (UAVs): A survey on civil applications and key research challenges," *IEEE Access*, vol. 7, pp. 48572–48634, 2019.
- [3] M. Campion, P. Ranganathan, and S. Faruque, "UAV swarm communication and control architectures: A review," *J. Unmanned Vehicle Syst.*, vol. 7, no. 2, pp. 93–106, Jun. 2019. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S2291346719000183>
- [4] J. Hu, A. Bruno, D. Zagieboylo, M. Zhao, B. Ritchken, B. Jackson, J. Yeon Chae, F. Mertil, M. Espinosa, and C. Delimitrou, "To centralize or not to centralize: A tale of swarm coordination," 2018, *arXiv:1805.01786*.
- [5] Y. Zhou, B. Rao, and W. Wang, "UAV swarm intelligence: Recent advances and future trends," *IEEE Access*, vol. 8, pp. 183856–183878, 2020.
- [6] İ. Bekmezci, O. K. Sahingoz, and Ş. Temel, "Flying ad-hoc networks (FANETs): A survey," *Ad Hoc Netw.*, vol. 11, no. 3, pp. 1254–1270, May 2013. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S1570870512002193>
- [7] K. Palan and P. Sharma, "Fanet communication protocols: A survey," *Int. J. Comput. Sci. Commun.*, vol. 7, pp. 219–223, Jan. 2015.
- [8] M. A. Khan, A. Safi, I. M. Qureshi, and I. U. Khan, "Flying ad-hoc networks (FANETs): A review of communication architectures, and routing protocols," in *Proc. 1st Int. Conf. Latest Trends Electr. Eng. Comput. Technol. (INTELLECT)*, Nov. 2017, pp. 1–9.
- [9] H. Chen and S. Megerian, "Cluster sizing and head selection for efficient data aggregation and routing in sensor networks," in *Proc. IEEE Wireless Commun. Netw. Conf.*, Apr. 2006, pp. 2318–2323.
- [10] P. S. Rathore, J. M. Chatterjee, A. Kumar, and R. Sujatha, "Energy-efficient cluster head selection through relay approach for WSN," *J. Supercomput.*, vol. 77, no. 7, pp. 7649–7675, Jul. 2021, doi: [10.1007/s11227-020-03593-4](https://doi.org/10.1007/s11227-020-03593-4).
- [11] H. Lu, J. Li, and M. Guizani, "Secure and efficient data transmission for cluster-based wireless sensor networks," *IEEE Trans. Parallel Distrib. Syst.*, vol. 25, no. 3, pp. 750–761, Mar. 2014.
- [12] S. Bandyopadhyay and E. J. Coyle, "An energy efficient hierarchical clustering algorithm for wireless sensor networks," in *Proc. 22nd Annu. Joint Conf. IEEE Comput. Commun. Societies*, Mar. 2004, pp. 1713–1723.
- [13] W. Su, A. K. Sadek, and K. J. Ray Liu, "Cooperative communication protocols in wireless networks: Performance analysis and optimum power allocation," *Wireless Pers. Commun.*, vol. 44, no. 2, pp. 181–217, Jan. 2008, doi: [10.1007/s11277-007-9359-z](https://doi.org/10.1007/s11277-007-9359-z).
- [14] D.-T. Ho, E. I. Grøtli, S. Shimamoto, and T. A. Johansen, "Optimal relay path selection and cooperative communication protocol for a swarm of UAVs," in *Proc. IEEE Globecom Workshops*, Dec. 2012, pp. 1585–1590.
- [15] J. Wu, L. Zou, L. Zhao, A. Al-Dubai, L. Mackenzie, and G. Min, "A multi-UAV clustering strategy for reducing insecure communication range," *Comput. Netw.*, vol. 158, pp. 132–142, Jul. 2019. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S1389128618312891>
- [16] T. M. Behera, S. K. Mohapatra, U. C. Samal, M. S. Khan, M. Daneshmand, and A. H. Gandomi, "Residual energy-based cluster-head selection in WSNs for IoT application," *IEEE Internet Things J.*, vol. 6, no. 3, pp. 5132–5139, Jun. 2019.
- [17] S. Tyagi and N. Kumar, "A systematic review on clustering and routing techniques based upon LEACH protocol for wireless sensor networks," *J. Netw. Comput. Appl.*, vol. 36, no. 2, pp. 623–645, Mar. 2013. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S1084804512002482>
- [18] A. Likas, N. Vlassis, and J. J. Verbeek, "The global k-means clustering algorithm," *Pattern Recognit.*, vol. 36, no. 2, pp. 451–461, Feb. 2003. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S0031320302000602>
- [19] L. Zheng, M. Ni, L. Cai, J. Pan, C. Ghosh, and K. Doppler, "Performance analysis of group-synchronized DCF for dense IEEE 802.11 networks," *IEEE Trans. Wireless Commun.*, vol. 13, no. 11, pp. 6180–6192, Nov. 2014.
- [20] G. Khraimech and F. Merazka, "Relay-based multiuser cooperative UAV networks over Rician channels," in *Proc. Int. Conf. Adv. Electron., Control Commun. Syst. (ICAECSS)*, Mar. 2023, pp. 1–6.
- [21] W. Tian, Z. Jiao, M. Liu, M. Zhang, and D. Li, "Cooperative communication based connectivity recovery for UAV networks," in *Proc. ACM Turing Celebration Conf.-China*, May 2019, pp. 1–5, doi: [10.1145/3321408.3321583](https://doi.org/10.1145/3321408.3321583).

- [22] P. Banerjee, "Study of cognitive radio performance in relation to signal to noise ratio and modulation scheme," *Int. J. Sci. Eng. Res.*, vol. 9, pp. 75–80, Jun. 2018.
- [23] A. K. Sarangi and A. Datta, "Capacity comparison of SISO, SIMO, MISO & MIMO systems," in *Proc. 2nd Int. Conf. Comput. Methodologies Commun. (ICCMC)*, Feb. 2018, pp. 798–801.
- [24] O. Chughtai, N. N. Qadri, Z. Kaleem, and C. Yuen, "Drone-assisted cooperative routing scheme for seamless connectivity in V2X communication," *IEEE Access*, vol. 12, pp. 17369–17381, 2024.
- [25] M. O. Abualhauja'a, S. Ansari, O. R. Popoola, L. Mohjazi, L. Bariah, S. Muhaidat, Q. H. Abbasi, and M. A. Imran, *RIS-Assisted UAV*. Piscataway, NJ, USA: IEEE Press, 2023, pp. 213–232. [Online]. Available: <https://ieeexplore.ieee.org/document/9966223>
- [26] P.-V. Mekikis, P. S. Bouzinis, N. A. Mitsiou, S. A. Tegos, D. Tyrovolas, V. K. Papanikolaou, and G. K. Karagiannidis, "Enabling wireless-powered IoT through incentive-based UAV swarm orchestration," *IEEE Open J. Commun. Soc.*, vol. 4, pp. 2548–2560, 2023.
- [27] S. Bhandari, X. Wang, and R. Lee, "Mobility and location-aware stable clustering scheme for UAV networks," *IEEE Access*, vol. 8, pp. 106364–106372, 2020.
- [28] B. Ginzburg and A. Kesselman, "Performance analysis of A-MPDU and A-MSDU aggregation in IEEE 802.11n," in *Proc. IEEE Sarnoff Symp.*, Apr. 2007, pp. 1–5.



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